

Supplementary Material: Data Quality for Derived Preference Graphs: Construction Sensitivity and Repair Outcomes in Multi-Ranker Retrieval

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This document is supplementary material for the main manuscript “Data Quality for Derived Preference Graphs: Construction Sensitivity and Repair Outcomes in Multi-Ranker Retrieval.” It is not self-contained: every section below expands a claim summarized in the compressed main paper, and no result here is presented as evidence independent of the main paper’s claims. Every table and figure in this document was computed by the same released, versioned pipeline as the main-text results (see the main paper’s Data Availability and Reproducibility section); nothing here is a new experiment. Sections A–H group the detail moved out of the main paper during the final page-limit compression.

1 A. EXTENDED PROTOCOL DEFINITIONS

Referenced from the main text’s “Threshold Protocols” subsection, which names four headline protocols (`raw_fixed`, `minmax_raw_matched`, `minmax_quantile`, `rank_percentile`) and states that, together with the $q \in \{0.3, 0.5, 0.7\}$ sensitivity grid, these correspond to eight registered protocol configurations. The joint multiplicity families additionally draw on four further registered configurations, defined here, bringing the total to the twelve referenced throughout the paper:

- `ablation_minmax_fixed` – `raw_fixed`’s fixed numeric thresholds (0.05/0.1), but with min–max calibration applied to scores before thresholding instead of raw native scores.
- `ablation_unit_vote_retention` – the same fixed numeric thresholds, but with unit-weight votes (each retained vote contributes weight 1) instead of margin-weighted votes.
- `robustness_zscore_retention` – constructed exactly like `minmax_raw_matched` (retention-matched to `raw_fixed`), but substituting per-query, per-ranker z-score standardization for min–max normalization.
- `robustness_rank_percentile_retention` – constructed exactly like `minmax_raw_matched`, but substituting the percentile transform for min–max normalization. Not to be confused with the independently-thresholded `rank_percentile` protocol, which uses the same percentile transform but a pre-registered quantile threshold rather than retention-matching.

These four are reported only inside the main paper’s joint multiplicity families (“Multiplicity and Influence Robustness” subsection) as an even more conservative robustness check; none is a headline protocol, and none changes the primary-protocol decision.

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Referenced from the main text’s “Normalization Changes Graph Construction” subsection. The main paper’s structural-results table already reports every value plotted below (cyclic-query percentage by dataset and regime, before and after mutual-pair deletion); the two figures here give the same values as grouped bar charts for visual comparison across datasets and against the raw-margin ablation.

Referenced from the main text’s “Ranker Dependence, Mutual-Pair Attribution, and Why ms2 Is Acyclic” subsection, which states in prose that the custom TF-IDF implementation was validated against scikit-learn’s `TfidfVectorizer` and matches closely.

TF-IDF implementation validation. We validated our TF-IDF implementation (described in the main paper’s “Upstream Rankers and Candidate Pooling” subsection) against scikit-learn’s `TfidfVectorizer`, configured to match its tokenizer, sublinear term frequency, smoothed IDF, and ℓ_2 -normalized cosine scoring, on a deterministic five-query sample per dataset scored against the full document corpus. Top-50 (35 for HotpotQA) overlap is exact (Jaccard = 1.0) on every sampled query, score Pearson correlation on jointly-ranked documents exceeds 0.9999, and directional pairwise agreement is 1.0 on every sampled query. This does not replace the stored canonical TF-IDF scores used throughout the paper; it confirms the implementation used to produce them is internally correct. Figures 1 and 2 visualize the structural cyclicity values summarized in the main paper.

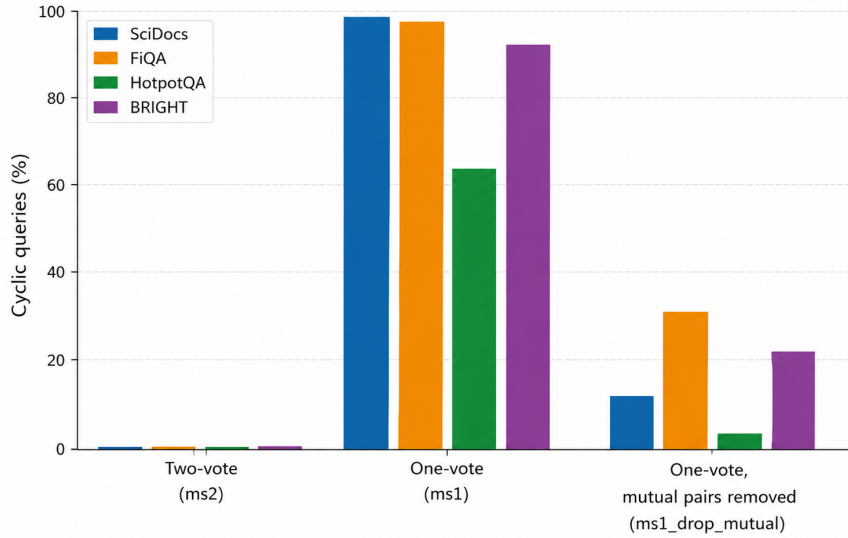


Fig. 1. Cyclic-query percentage under the primary normalized protocol, grouped by vote regime with one bar per dataset. ms2 is exactly 0% on every dataset; ms1 is 64–99%; deleting direct mutual pairs (ms1_drop_mutual) brings every dataset back down to 2–31%, since ms1’s high cyclicity is concentrated almost entirely in direct mutual contradictions rather than longer cycles (the main paper’s mutual-pair-deletion figure isolates that residual). Values before mutual-pair deletion; post-deletion values are also in the main paper’s structural-results table.

2 B. FULL STATISTICAL ROBUSTNESS

Referenced from the main text’s “Multiplicity and Influence Robustness” subsection, which states the exact per-step influence-removal values for both panels below in prose (HotpotQA: $+0.0123 \rightarrow +0.0053 \rightarrow +0.0009 \rightarrow 0 \rightarrow 0$ as the

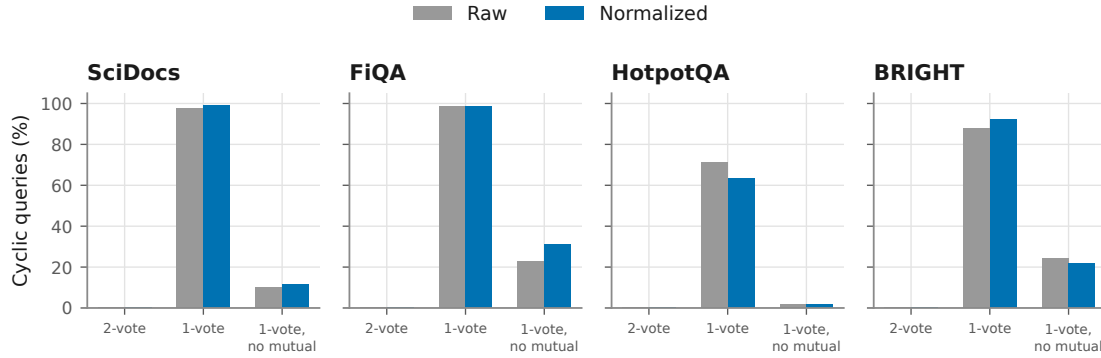


Fig. 2. Cyclic-query percentage by vote regime, raw-margin ablation (gray) versus primary normalized protocol (blue), one panel per dataset. Unlike the BM25 weight share in the main paper’s scale-dominance figure, overall cyclic-query rate changes only modestly between raw and normalized construction within each regime; this does not mean normalization leaves the graph unchanged, since the removed-edge overlap between raw and normalized repair is only 0.055–0.328, so normalization changes *which* edges are cyclic and removed far more than it changes the headline cyclic-query rate shown here.

top 1-4 influential queries are removed; SciDocs (at $k = 0, 1, 2, 3, 4$): $+0.0085 \rightarrow +0.0055 \rightarrow +0.0036 \rightarrow +0.0048 \rightarrow +0.0041$).

Referenced from the main text’s “Alpha Sensitivity and Balance Degeneracy” subsection, which states in prose which datasets/ α combinations remain positive, negative, or sign-flipping across the sweep. Table 1 gives the full alpha grid, and Figure 3 shows the corresponding query-influence sensitivity traces cited in the main text.

Table 1. Copeland-hybrid alpha sensitivity under the primary normalized protocol, ms1 only. Entries are repaired-minus-unrepaired mean nDCG@ k ; this is a small sensitivity sweep, not a tuned hyperparameter search.

Dataset	$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 1.0$
SciDocs	−0.0022	+0.0085	+0.0097	+0.0083
FiQA	+0.0006	−0.0046	+0.0010	+0.0038
HotpotQA	−0.0008	+0.0123	+0.0120	+0.0101
BRIGHT	−0.0013	+0.0006	−0.0074	−0.0071

3 C. EXACT REPAIR ANALYSES

Referenced from the main text’s “Structural Repair Activity” subsection, which states the same finding in prose: ms2 removes zero weight on every dataset by construction, ms1 removes 0.029–0.080 of total graph weight, and ms1_drop_mutual removes only a small residual (≤ 0.004). Figure 4 gives the full per-dataset decomposition used by that summary.

4 D. CANDIDATE-POOL AND CUTOFF ROBUSTNESS

Referenced from the main text’s “The baseline comparison is not an artifact of the RRF-centered candidate pool” paragraph, which states the headline result in prose: under the canonical pool, 0 of 8 pre-specified Holm-corrected cells

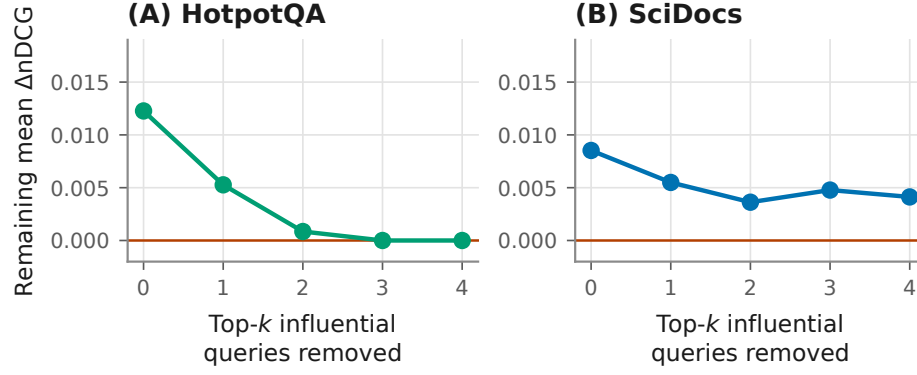


Fig. 3. Remaining mean $\Delta nDCG$ for the normalized ms1 Copeland-hybrid cell as the top- k most influential queries are removed ($k = 0, \dots, 4$), exact per-query recomputations, not interpolated. HotpotQA (A) is the more concentrated case: the mean falls from +0.0123 to exactly 0 after removing the top 3 queries. SciDocs (B) is less concentrated and stays positive through $k = 4$ (+0.0085 at $k = 0$, never dropping below +0.0036).

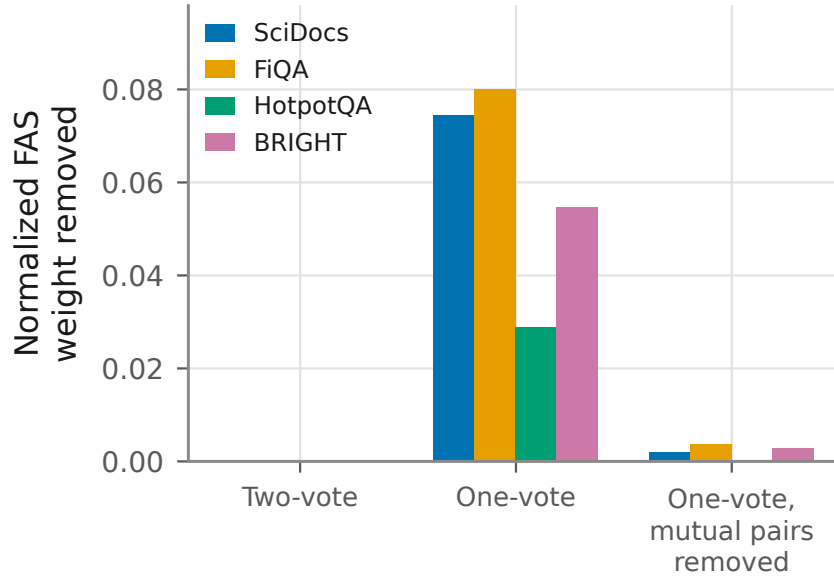


Fig. 4. Mean normalized FAS weight removed (removed weight divided by total graph weight) by vote regime, one bar per dataset, under the primary normalized protocol. ms2 removes exactly 0 on every dataset by construction (it is always acyclic); ms1 removes 0.029–0.080 of total graph weight; ms1_drop_mutual removes only a small residual (≤ 0.004). This is an intrinsic graph diagnostic computed before any retrieval evaluation, not a measure of retrieval benefit.

are significant, and the one nominally significant cell across all five pools tested occurs under the neutral round-robin pool, not the canonical one – the opposite of a home-field-advantage pattern.

Referenced from the main text’s “Conditional Analysis and Failure Decomposition” subsection, which restates every value below in prose while walking through the nested-subset funnel for the HotpotQA ms1 Copeland-graph cell at $(P, k) = (35, 10)$. Tables 2 and 3 hold the moved conditional and pool-fairness detail.

Table 2. Conditional decomposition of the HotpotQA ms1 Copeland-graph cell at $(P, k) = (35, 10)$ (52 queries total). Sample size and mean $\Delta nDCG$ for each nested subset.

Subset	n	Mean $\Delta nDCG$
All queries	52	+0.0165
Has a cycle / repair active	44	+0.0195
Repaired ranking differs from unrepaired	43	+0.0200
Top- k document set changes	19	+0.0219
Qrels-labeled document ordering changes	8	+0.0854

Table 3. Targeted paired comparison of {RRF, CombSUM} vs. the fixed repaired Copeland hybrid (hybrid_repaired_copeland_a0p3_minmax), ms1 regime, mean nDCG delta (baseline minus graph method). The primary canonical-pool family (8 cells) is Holm-corrected jointly and is the only family used for an inferential claim; the neutral-pool column is reported for robustness context only. Full five-pool results are in the released artifact.

Dataset	Baseline	Canonical (RRF) pool		Neutral pool	
		$\Delta nDCG$	Holm p	$\Delta nDCG$	Holm p
SciDocs	RRF	−0.0066	0.625	−0.0058	0.863
SciDocs	CombSUM	−0.0099	0.625	−0.0051	1.000
FiQA	RRF	+0.0141	0.058	+0.0082	0.629
FiQA	CombSUM	+0.0135	0.342	+0.0140	0.013
HotpotQA	RRF	−0.0197	0.625	−0.0103	1.000
HotpotQA	CombSUM	+0.0045	0.625	+0.0127	1.000
BRIGHT	RRF	+0.0137	0.342	+0.0043	1.000
BRIGHT	CombSUM	+0.0250	0.198	+0.0123	1.000

5 E. RANKER DEPENDENCE AND COVERAGE

Referenced from the main text’s “Structural Repair Activity” subsection, which states in prose that, under the principled judged-pair rule, SciDocs and HotpotQA provide usable qrels-pair diagnostics while FiQA and BRIGHT do not in the analyzed candidate pools.

Referenced from the main text’s “Ranker Dependence...” subsection, which states in prose that pre-pool and post-pool normalization produce nearly identical graphs. Table 4 reports the qrels-pair availability diagnostics used to support that scope statement.

Pre-pool versus post-pool normalization order. The primary protocol normalizes each ranker’s scores after restricting to the candidate pool. We compare this to normalizing over each ranker’s full stored top- n score list first and only then restricting to the pool, using an independently-defined median-margin threshold for both constructions (declared before inspecting results) so that neither construction is favored by a reused retention target from the other. The two constructions produce nearly identical edge counts, densities, and mutual-pair counts (e.g. SciDocs ms1 canonical pool: 153.9 vs. 154.1 mean edges), removed-edge Jaccard overlap between the two constructions is 0.748–1.00 across every dataset/pool/regime cell checked, and a pre-specified ms1-only active-family Holm correction finds 0 of 32 cells significant. Candidate-pool size still changes graph structure substantially, but that change is attributable to the change in candidate-set composition itself, not to an interaction between pool restriction and normalization order.

Table 4. Availability of qrels-pair diagnostics under the canonical small-pool evaluation design. “Eligible pairs” count candidate pairs for which both documents have explicit qrels and different relevance grades; equal-grade pairs and pairs containing unjudged documents are excluded. A zero rate of 1.0 means the qrels-pair diagnostics are unavailable on every analyzed query for that dataset/pool.

Dataset	Mean judged docs	Mean eligible pairs	Zero-pair query rate
SciDocs	29.93	123.33	0.00
FiQA	1.87	0.00	1.00
HotpotQA	9.83	10.85	0.02
BRIGHT	5.58	0.00	1.00

6 F. ADDITIONAL BASELINE DETAILS

Referenced from the main text’s “Methods and Baselines” subsection, which states that PageRank, RankCentrality, and Bradley–Terry were added to cover three methodologically distinct graph-ranking families not already represented by Copeland, balance, and Markov extraction.

Why these three families, and not others. PageRank is a damped random-walk/authority-propagation method that computes weighted authority on the reversed preference graph [3]; RankCentrality [2] is a damping-free stationary-distribution estimator that is a known consistent approximation of the Bradley–Terry maximum-likelihood solution even on cyclic comparison graphs, representing the random-walk family without a teleportation term; Bradley–Terry [1] is a probabilistic pairwise-comparison model that fits per-document strength parameters by maximum likelihood via iterative scaling directly on the graph’s weighted edges, treated as pairwise win/loss counts, representing a fundamentally different (likelihood-based rather than combinatorial or spectral) ranking principle. HodgeRank, Elo, and TrueSkill were not added: a Helmholtz/harmonic decomposition of pairwise comparisons, or an online rating update rule, would extend the same probabilistic-pairwise or spectral families already covered by Bradley–Terry and RankCentrality respectively, so they are not a scientifically required addition for this study’s comparison, only a possible future extension.

Referenced from the main text’s “Comparison with Fixed Aggregation Baselines” subsection, which reports the same information in its dataset-macro comparison table and states its practical takeaway (a graph-independent baseline is within the top three methods on every dataset) in prose. Figure 5 shows the full per-dataset ordering, and Figure 6 shows the raw-versus-normalized sign-instability grid.

Referenced from the main text’s “Raw Versus Normalized Sign Instability” subsection, which reports the five exact sign-flip values in its own table and states the aggregate finding (30% of dataset×regime×pair cells lack a consistent sign across the three headline protocols, none individually significant) in prose.

7 G. LLM PROTOCOL AUDIT

Referenced from the main text’s “Protocol Audit of Stored LLM Judgments” subsection, which states the headline finding in prose (removing parser-default and position-bias-driven edges reduces measured cyclicity by 91–100%, from a 31–62% cyclic range down to 0–5%). The table below adds the two supporting per-corpus statistics not otherwise stated in the main text: the overall parser default-label rate and the forward/reverse prompt-order agreement rate. Table 5 is the moved quantitative summary.

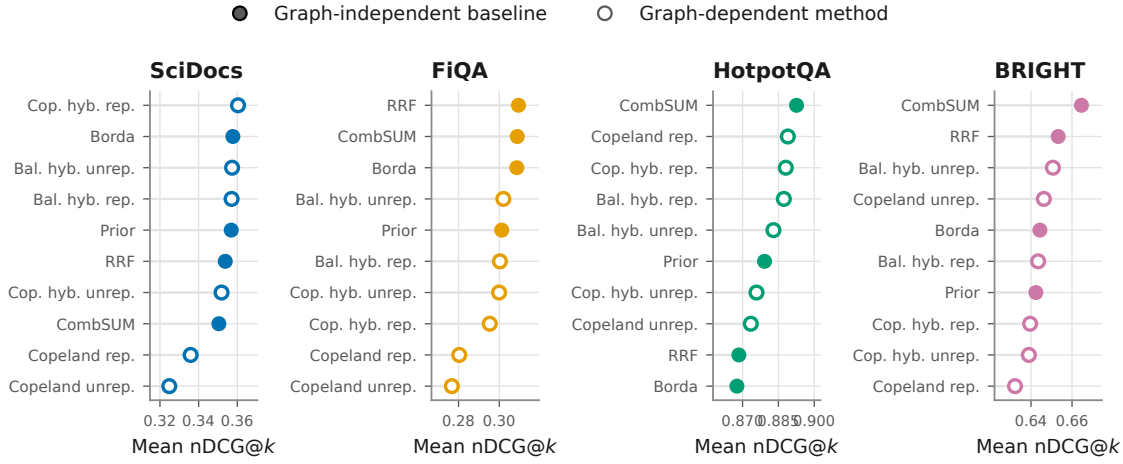


Fig. 5. Mean nDCG@k under ms1, one panel per dataset, methods on the y -axis sorted by value; filled markers are the four graph-independent baselines (Prior, RRF, CombSUM, Borda), open markers are graph-dependent methods. Each panel's x -axis is scaled to its own score range, so only within-panel comparisons are meaningful. A graph-independent baseline is within the top three methods on every dataset (CombSUM leads on HotpotQA and BRIGHT, RRF on FiQA), while on SciDocs a repaired Copeland-hybrid edges out CombSUM narrowly – consistent with the main paper's macro-average ranking table, which pools across all four datasets.

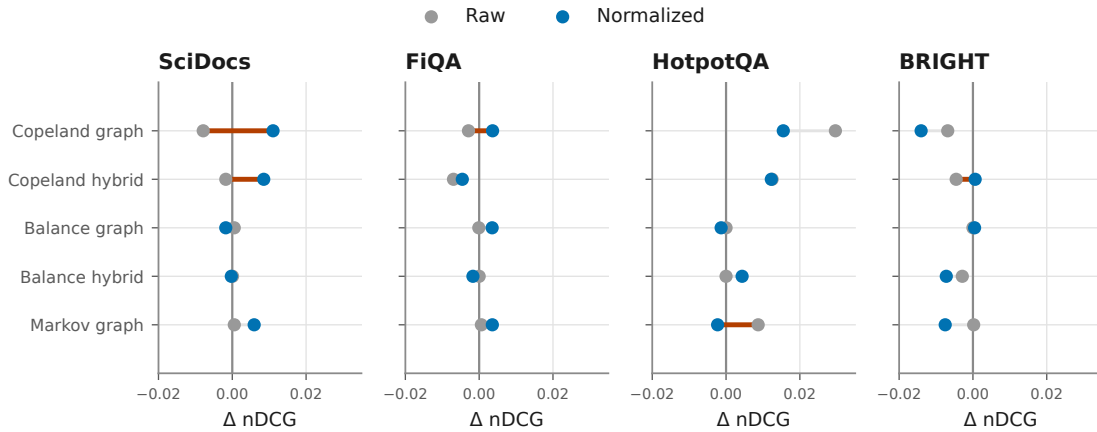


Fig. 6. Repaired-minus-unrepaired Δ nDCG for every method pair under ms1, one panel per dataset: gray dot is the raw-margin ablation, blue dot is the primary normalized protocol, joined by a line, against a zero reference. The five accent-colored lines are the sign flips the main paper's raw-vs-normalized sign-flip table reports exactly; other pairs keep the same sign or flip only at noise-level magnitude on both sides.

Table 5. Quantitative protocol audit of the stored real-LLM judgment corpora (Cohere and Azure OpenAI, FiQA/HotpotQA/BRIGHT, 12,020 responses). Provider-specific detail beyond this summary is in the released artifact.

Quantity	Value
Parser default-label rate	1.1% overall (0.0%–3.8% by provider/dataset)
Position bias, Azure	prefers 1st-shown, 53–58%
Position bias, Cohere	prefers 2nd-shown, 61–70%
Forward/reverse agreement	59–85% (Cohen’s κ 0.27–0.71)
Cyclicity after artifact removal	–91% to –100% (31–62% $\rightarrow$ 0–5%)

8 H. FULL REPRODUCIBILITY TABLES

Every canonical aggregate table, protocol manifest, and statistical-family definition referenced throughout this supplement is released in full in the accompanying reproducibility artifact (see the main paper’s Data Availability and Reproducibility section); this section does not repeat that inventory.

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